

No. of Functional Features Included in the Solution

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Project Name	EcoTrack - AI-Powered Carbon Footprint Tracker
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the EcoTrack solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Daily Habit Logger	Multi-step form to log travel, food, and energy activities for the day.	Frontend (React.js)	Done	1.0
2	Firebase Authentication	Secure user sign-up and login using Firebase Auth.	Security / Auth	Done	0.5
3	Carbon Emission Calculator	Calculates travel, food, and energy emissions using real-world factors from Carbon Interface API.	Backend (Flask)	Done	1.0
4	AI Eco-Suggestion Engine	Generates 3 personalised eco-friendly lifestyle recommendations using Groq LLaMA 3.3-70B.	AI / Backend	Done	1.0
5	Animated Carbon Gauge	Visual gauge showing daily footprint against the personal carbon budget.	Frontend (React.js)	Done	0.5
6	Trend Dashboard	30-day line chart, 14-day stacked bar chart, and heatmap calendar for footprint history.	Frontend (Chart.js)	Done	0.5
7	Personal Carbon Budget Tracker	Allows users to set and track progress against a configurable daily carbon budget.	Backend / Firestore	Done	0.25
8	Streak & Badge Achievement System	Tracks consistency streaks and awards badges for sustained eco-friendly behaviour.	Backend / Gamification	In Progress	0.25

Feature Summary:

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Daily Habit Logger, Animated Carbon Gauge, Trend Dashboard
2	Backend / Logic	3	Carbon Emission Calculator, Carbon Budget Tracker, Streak & Badge System
3	Database / Storage	1	Firebase Firestore (habit logs, badges, user settings)
4	API / Integration	2	Carbon Interface API, Groq AI (LLaMA 3.3-70B)
5	Security / Authentication	1	Firebase Authentication (sign-up / login)